



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	7.40 / 10
2022	MSBSHSE(XII)	Rajarshi Shahu Junior Science College, Latur	82.00%
2020	MSBSHSE(X)	Sri Sri Ravishankar Madhyamik Vidyalaya, Latur	98.20%

INTERNSHIPS

FlyRank AI | Backend AI Engineering Intern Jul '26 - Aug '26

- Architected a **multi-tenant billing system** with a **5-table relational schema**, ensuring isolation and consistency across tenants at scale.
- Built a **scalable async metering pipeline** using **SQLAlchemy** to process usage events into integer-precise cost rollups, preventing drift.
- Eliminated duplicate billing via dual-layer **idempotency** (idempotency keys + **DB constraints**) under concurrent webhook retries scenarios.
- Integrated Stripe-style **resilient webhook processing** with idempotent handlers and retry-safe logic, ensuring **exactly-once processing**.

Trinity | AI Engineering Intern Jun '25 - Aug '25

- Engineered a **production-grade RAG system** using hybrid retrieval (FAISS + BM25) and reranking, achieving **90%+ faithfulness** across 60 queries.
- Improved retrieval performance by boosting **Recall@5** from **~68% to ~91%** via hybrid search and reranking, reducing retrieval noise.
- Optimized **inference efficiency** by reducing context from 20 to 4 chunks via reranking, cutting **LLM cost per query by ~60%** without accuracy loss
- Reduced system latency by **~30% (p95)** using ANN indexing (**HNSW**) and reranker pruning, enabling scalable low-latency retrieval

PROJECTS

CodeAgentPro | Multi-Agent LLM System with RAG, Evaluation & Sandbox Execution | Self Project Mar '26 - Apr '26

- Python, FastAPI, LangGraph, LLM (Groq API, Gemini API, Ollama), ChromaDB, Sentence-Transformers, Docker, Server-Sent Events (SSE)*
- Orchestrated **multi-agent** LangGraph pipeline with adaptive **self-repair** (code, test, debug via RAG memory), enabling autonomous recovery
 - Optimized RAG system with **ChromaDB cache** across 3 vector stores, reducing lookup latency by **99.8%** and improving **retrieval efficiency**
 - Devised multi-language test harness (Pytest, Node, C++) evaluating Pass@1 across 18-task benchmark via GitHub Actions CI pipelines
 - Unified **multi-LLM providers** (Groq, Gemini, Ollama) with **async streaming** and **Docker sandboxing**, enabling real-time agent execution runtime.

P2P File Sharing System | Self Project Jan '26 - Feb '26

- C, Socket Programming(TCP/IP), POSIX Threads, P2P System Architecture, Mutes Synchronization*
- Built a **multi-threaded** server with **POSIX threads** and **mutexes** to handle peer registration, file queries, and concurrent client requests
 - Developed a **dual-function** client that registers files, hosts downloads, and fetches files dynamically from peers using **TCP connections**
 - Ensured **thread-safe access** to shared peer lists and metadata, supporting up to **10 concurrent** peer nodes reliably for file operations
 - Designed a cross-platform **Makefile** with OS detection and **install/uninstall** rules for Linux, macOS, and Windows via MSYS2 environments

Expense Splitter & Cashflow Minimizer | Self Project Oct '25 - Nov '25

- C++, Object-Oriented Design, System Architecture*
- Designed a modular C++ expense engine supporting equal, exact and percentage splits using **Strategy**, **Factory** and **Facade patterns**.
 - Reduced inter-user settlement by **90%** by implementing a **greedy netting algorithm** to simplify debt graphs and remove redundant payments.
 - Integrated an **Observer-pattern** notification engine delivering real-time balance updates and transaction alerts to **50+ group members**

COMPETITION/CONFERENCE

General Championship Data Analytics 2024 | IIT Kharagpur & Evva Health Feb '24 - Mar '24

- Secured **Gold** in the General Championship Data Analytics with a hall team, outperforming 15+ top-performing teams at IIT Kharagpur
- Curated 3 years of SDOH datasets across Census tract, ZIP, and county levels, merging them via crosswalk mapping into one dataset
- Mapped SDOH features to **11 risk categories** using **Sentence-BERT** and **cosine similarity**, then clustered variables within risk using **K-Means**
- Built a percentile-based thresholding algorithm and deployed a Streamlit Risk Calculator app recommending personalized health support

AWARDS AND ACHIEVEMENTS

- Solved **1500+ DSA problems** across LeetCode, CodeChef, CSES and Codeforces demonstrating strong algorithmic problem-solving skills.
- Achieved **1770 LeetCode (nikhilmahi)** and **3 star CodeChef (1662, nikhilmahi)**, consistently performing across contests and practice problems.

SKILLS AND EXPERTISE

Programming Languages : Python, C++, SQL, C, JavaScript **Backend and Systems** : FastAPI, Rest APIs, AsyncIO, SQLAlchemy, System Design
Databases and Storage : PostgreSQL, Redis, SQLite **Tools & Infrastructure** : Docker, Docker Compose, Git, GitHub, Linux, n8n
Core CS : Data Structures and Algorithms, Object Oriented Programming System, Operating Systems

COURSEWORK INFORMATION

Coursework : Linear Algebra | Numerical and Complex Analysis | Advanced Calculus | Programming and Data Structures* | Economics
MOOCs : Machine Learning | Deep Learning | Generative AI | Agentic AI | Probability and Statistics (*courses with lab)

EXTRA CURRICULAR ACTIVITIES

- Led LLR Hall as **General Secretary** to podium finishes in GC Sports and Games 2025 with **Silver** in Chess and **Bronze** in Water Polo and Tennis.
- Represented the Hall in the General Championship and secured **Gold in Water Polo (2024)** and achieved **4th place in Cricket (2026)**.